A five-count comparison with one proof

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The same argument treats the two conditional support strips and a narrower strip supplied by geometry. Five neighboring counts retain a definite crossing loss. Four centered replacements estimate that loss at variance cost, while one fourth moment controls all omitted translates. The remaining binomial and normal corrections have order $1/n$. Their final comparison is a pair of concave quartics, so only four endpoint inequalities are needed for each row of constants.

Write $\phi_0 = (2\pi)^{-1/2}$ and $C_{\mathrm{E}} = (3+\sqrt{10})\phi_0/6$. Let $Y = B + U$, where B is Bernoulli of parameter p and $\mathbb{E}(U \mid B) = 0$. Denote the conditional laws by U_0, U_1, and put

$$q = 1-p, \quad v = pq, \quad s_j = \mathbb{E}U_j^2, \quad s = qs_0 + ps_1, \quad \lambda = ns, \quad X = \frac{Y-p}{\sqrt{v+s}}, \quad \rho = \mathbb{E}|X|^3.$$

In every statement below the noise laws are those of this one marginal.

Theorem 1. *Choose any row of Table 1. Suppose $n \ge N$, $p \in [p_-, p_+]$, $L \le \lambda \le U$, and*

$$-A \le U_0 \le a_0, \qquad -a_1 \le U_1 \le A, \qquad a_0, a_1 \ge 0, \qquad a_0 + a_1 \le d.$$

For independent copies $X_1, \dots, X_n$, and every $|H| \le 46/25$,

$$\sqrt{n}\{\mathbb{P}(X_1 + \cdots + X_n \le H) - \Phi(H/\sqrt{n})\} - C_{\mathrm{E}}\rho < -\lambda/100. \tag{1}$$

Row	N	$[p_-, p_+]$	A	d	L	U
Compact	700	$[3/8, 93/200]$	$3/5$	$1/250$	$1/100$	$7/20$
Right	200	$[2/5, 53/100]$	$3/5$	$9/125$	$3/100$	$9/25$
Left	700	$[1/3, 2/5]$	$7/10$	$17/250$	$1/100$	$8/25$

Table 1: Three specializations of the same criterion.

The compact row is the natural comparison after the narrower geometric acquisition. The two other rows remain useful without that acquisition. In particular the right row now starts at $\lambda = .03$, so its smaller-noise join needs only the existing all-phase .03 comparison. Each displayed theorem is local: acquiring its support and raw threshold, reflection and an independent finite base remain separate obligations.

1 A criterion with explicit parameters

Here is the common criterion used to prove the theorem. Assume

$$N \ge 42, \quad [p_-, p_+] \subset [1/3, 2/3], \quad 0 < L \le U, \quad 0 \le d \le A, \quad d < r, \quad A + d < 1,$$

where

$$r = \min(p_-, 1-p_+), \quad q_* = 1-p_+, \quad v_- = \min_{p_- \le p \le p_+} p(1-p), \quad v_+ = \max_{p_- \le p \le p_+} p(1-p).$$

The following constants are fixed throughout the proof:

$$\eta = 2/5, \quad \beta = 41/100, \quad \pi_+ = 22/7, \quad \phi_- = 199/500, \quad \phi_+ = 2/5.$$

Choose positive $\kappa, \xi, c_-, \zeta, m, V, j_0, j_1$, and $\bar{c} \in [0,1)$, satisfying these conditions:

$$\eta\pi_+^2 d \le \frac{1-A-d}{A+d}, \qquad 1-\cos(2\pi x) \ge \kappa x^2 \quad (|x| \le A+d), \tag{2}$$

$$(46/25)^2(v_+ + U/N) < \xi^2, \quad \xi < 1, \quad N(r-d) > \xi, \quad c_- r \ge 1+\xi, \quad c_- < N, \tag{3}$$

$$0 < \zeta < \min_{z=p_- -4/N,\ p_+ +4/N} z(1-z), \qquad 0 < m < 2\phi_- \left(1 - \frac{8+1/12}{N\zeta}\right), \tag{4}$$

$$V^2 < v_-^3, \qquad \bar{c} \ge (\xi - q_*)_+, \qquad 4j_0 j_1 \ge 1/q_*, \tag{5}$$

$$K_0 - K_1 U/N > 0, \quad K_0 = 3r - d - \tfrac{3}{2}(1-2v_-), \quad K_1 = \frac{3(1-2v_-)}{8v_-}. \tag{6}$$

All divisions are by positive quantities. The points in (4) lie in $(0,1)$ because $N \ge 42$ and $[p_-, p_+] \subset [1/3, 2/3]$.

Put $\alpha = 1 - c_-/N$, $b = 1 + 3/N$, $z = \kappa\alpha$, and define

$$W(\lambda) = \eta m \left\{ \frac{z\lambda - z^2\lambda^2/2 + z^3\lambda^3/6 - z^4\lambda^4/24}{2} - \frac{A^2 b\lambda + 3b^2\lambda^2}{16} - \frac{4\pi_+^2\lambda}{Nr} \right\}. \tag{7}$$

The common error polynomials are

$$T = \lambda/q_*, \quad M = A^2 T + 3T^2, \quad J(\lambda) = j_1\lambda + j_0, \quad S_c = \frac{T}{4} + cJ(\lambda) + \frac{27M}{256}\left(\frac{5}{3} + \frac{5c}{4}\right),$$

$$E_- = \frac{\beta S_{\bar{c}} + \phi_+(\xi/8 + 2/27)}{V}, \qquad E_+ = \frac{\beta S_0 + \phi_+(2/27 + \xi\lambda/2)}{V}. \tag{8}$$

Lemma 2 (Four endpoint criterion)**.** *Under* (2)–(6)*, conclusion* (1) *follows if the four rational quantities*

$$P_\pm(L), P_\pm(U) \quad \textit{are positive}, \qquad P_\pm(\lambda) = W(\lambda) - E_\pm(\lambda)/N - \lambda/100.$$

The proof below supplies the reduction, rather than assuming that endpoint checks enclose a continuous domain.

2 One exact crossing identity

Set $t = np + H\sqrt{v+s}$, $k = \lfloor t \rfloor$, $u = t - k \in [0,1)$ and $w = t - np$. By (3), $|w| < \xi$ and $na_0 \le t < n(1-a_1)$. On one auxiliary space take independent copies of the conditional laws and write

$$T_j = j + \sum_{l \le j} U_{1l} + \sum_{l > j} U_{0l}, \quad L_j(t) = \mathbb{E}(t - T_j)_+, \quad V_i = \sum_{l < i} U_{1l} + \sum_{l > i} U_{0l}.$$

The mesh increases because $T_j - T_{j-1} \ge 1 - a_0 - a_1 > 0$. Every T_j has exactly the conditional convolution of the actual iid sample given j successes. Define

$$h(x) = \mathbb{E}(x - U_0)_+ - \mathbb{E}(x - 1 - U_1)_+ - \mathbb{P}(1 + U_1 \le x), \qquad g_i = \mathbb{E}h(t - i + 1 - V_i).$$

Independence of the omitted coordinate gives

$$g_i = L_{i-1}(t) - L_i(t) - \mathbb{P}(T_i \le t).$$

Here $L_0 = t$, $L_n = 0$, and $\mathbb{P}(T_0 \le t) = 1$. Multiplication by the binomial masses π_i and finite summation by parts therefore yields

$$\mathbb{P}\Big(\sum X_i \le H\Big) = (1-u)F_B(k) + uF_B(k+1) + \sum_{j=1}^{n-1}(\pi_{j+1} - \pi_j)B_j - \sum_{i=1}^{n}\pi_i g_i, \tag{9}$$

where $B_j = L_j(t) - (t-j)_+ \ge 0$ by Jensen. To verify the interpolation term, replace every L_j in $\pi_0 + \pi_1 t + \sum_{j=1}^{n-1}(\pi_{j+1} - \pi_j)L_j$ by $(t-j)_+$ and telescope once more. All CDF events in these formulas are closed. The strict survival events used next are complementary to precisely these closed events.

3 Five kernels, four centered replacements

Let $f(x) = \sin^2(\pi x)$ on $[0,1]$ and zero elsewhere. It is continuously differentiable, its derivative vanishes at the joins, and that derivative is Lipschitz with constant $2\pi^2$. We first prove the common pointwise inequality

$$h(x) \ge \eta f(x) \ge 0. \tag{10}$$

For $x < 1 - a_1$, $h(x) = \mathbb{E}(x - U_0)_+ \ge 0$, and for $x \ge 0$ Jensen improves this to $h(x) \ge x$. On $[0,1]$ one has $f(x) \le (5/2)x$: use $\pi^2 < 10$ up to $1/4$, the concave tangent $1/2 + \pi(x - 1/4)$ on $[1/4, 2/5]$, and $f \le 1$ thereafter. The two tangent endpoint inequalities follow from $\pi < 10/3$.

For $x \ge 1 - a_1 > a_0$, put $D = 1 + U_1$ and $L = \mathbb{E}(D - x)_+$. Put–call parity gives $h(x) = \mathbb{P}(D > x) - L$. Since $D \le 1 + A$, this is nonnegative: it is at least $(x - A)\mathbb{P}(D > x)$ until $x = 1 + A$ and is zero beyond that point. If $1 - a_1 \le x \le 1$, then $L \ge 1 - x$ and

$$h(x) \ge \frac{x - A}{1 + A - x}L \ge \frac{1 - A - d}{A + d}(1 - x) \ge \eta\pi^2(1 - x)^2 \ge \eta f(x).$$

The comparisons at $x = 0, 1$ are weak, as both sides may vanish. In particular all atoms of D are included.

Freeze $V = V_{k+1}$ and let $V^{(j)}$ have the law of V_{k+1+j}. There are $n - 1$ summands in each. For $j > 0$, the latter law replaces j copies of U_0 in V by U_1. Put $f_j(x) = f(u - j - x)$. The two positive changes regroup as

$$\begin{aligned}&\mathbb{E}f_1(V^{(1)}) + \mathbb{E}f_2(V^{(2)}) - \mathbb{E}f_1(V) - \mathbb{E}f_2(V)\\ &= \mathbb{E}(f_1 + f_2)(V^{(1)}) - \mathbb{E}(f_1 + f_2)(V) + \mathbb{E}f_2(V^{(2)}) - \mathbb{E}f_2(V^{(1)}).\end{aligned}$$

The negative changes have the same form with the laws interchanged. The translated bumps have disjoint open supports and zero derivatives at every join. Thus each function in these four replacements still has derivative Lipschitz constant $2\pi^2$.

For a centered Z, Taylor's integral remainder gives $|\mathbb{E}F(a + Z) - F(a)| \le \frac{1}{2}\operatorname{Lip}(F')\mathbb{E}Z^2$. Condition on the other independent summands and apply this twice to each replacement. The total cost is at most $4\pi^2(s_0 + s_1)$, so

$$\sum_{i=k-1}^{k+3} g_i \ge \eta\left\{\sum_{j=-2}^{2}\mathbb{E}f(u - j - V) - 4\pi^2(s_0 + s_1)\right\}.$$

The five indices are interior: $|i - np| < \xi + 3 < 4$. No independence of sums at different counts is used.

The infinite translates of f sum to $\sin^2(\pi(u - V))$. An omitted nonzero translate with $|j| \ge 3$ requires $|V| > 2$. The bump vanishes at both endpoints, so this remains true at collisions. Consequently

$$\sum_{j=-2}^{2}\mathbb{E}f(u - j - V) \ge \mathbb{E}\sin^2(\pi(u - V)) - \mathbb{E}V^4/16.$$

Writing $\delta = k - np \in (-1-\xi, \xi)$, the actual frozen variance is

$$\tau = (nq - \delta - 1)s_0 + (np + \delta)s_1.$$

The phase guards and $r \geq 1/3$ give

$$(1 - c_-/n)\lambda \leq \tau \leq (1 + 3/n)\lambda, \quad s_0 + s_1 \leq \lambda/(nr), \quad \mathbb{E}V^4 \leq A^2\tau + 3\tau^2.$$

The last bound follows by expanding the fourth moment of the independent centered summands, using $\mathbb{E}U_j^4 \leq A^2 s_j$.

For either noise Z, an independent copy Z' satisfies $|Z - Z'| \leq A + d$. The cosine guard implies

$$1 - |\mathbb{E}e^{2\pi i Z}|^2 = \mathbb{E}\{1 - \cos(2\pi(Z - Z'))\} \geq 2\kappa \operatorname{Var} Z.$$

Hence $|\mathbb{E}e^{2\pi i Z}| \leq e^{-\kappa \operatorname{Var} Z}$. Multiplying only over the independent frozen summands proves

$$\mathbb{E}\sin^2(\pi(u - V)) \geq (1 - e^{-\kappa\tau})/2.$$

We have obtained the generic five-count reserve

$$\sum_{i=k-1}^{k+3} g_i \geq \eta \left\{ \frac{1 - e^{-\kappa(1-c_-/n)\lambda}}{2} - \frac{A^2(1 + 3/n)\lambda + 3(1 + 3/n)^2\lambda^2}{16} - \frac{4\pi^2\lambda}{nr} \right\}. \tag{11}$$

4 Finite binomial weights and curvature

We use finite Stirling with its explicit remainder:

$$\log(j!) = (j + \tfrac{1}{2})\log j - j + \tfrac{1}{2}\log(2\pi) + r_j, \qquad 0 < r_j < 1/(12j).$$

For clarity, if $x = 1/(2j+1)$ then $r_j - r_{j+1} = \operatorname{arctanh}(x)/x - 1 = \sum_{\ell \geq 1} x^{2\ell}/(2\ell + 1)$ lies strictly between zero and $1/(12j(j+1))$. The differences are summable, and Wallis's product fixes the resulting constant, so $r_j \to 0$. Summing the differences from j to infinity gives the displayed bound. No local central limit theorem or law-dependent asymptotic remainder is used.

For $k - 1 \leq i \leq k + 3$ and $z = i/n$, the segment between z and p has variance product at least ζ. Binary relative entropy therefore satisfies $nD(z\|p) \leq (i - np)^2/(2n\zeta) < 8/(n\zeta)$. The factorial remainder is bounded below by $-1/(12n\zeta)$. Since $z(1 - z) \leq 1/4$,

$$\sqrt{n}\,\pi_i > 2\phi_- \left(1 - \frac{8 + 1/12}{N\zeta}\right) > m.$$

All other g_i are nonnegative. The exact reserve (11) increases with n at fixed λ. First replace n by N there, then use $1 - e^{-x} \geq x - x^2/2 + x^3/6 - x^4/24$ for $x \geq 0$. This proves

$$\sqrt{n} \sum_i \pi_i g_i \geq W(\lambda). \tag{12}$$

The order of these steps does not assume monotonicity of a quartic approximation in n.

We give the full summable curvature estimate once. Put $c = (np - q - t)_+$. If T, M bound $\mathbb{E}W_j^2, \mathbb{E}W_j^4$ on the increasing side of the binomial mass, where $W_j = T_j - j$, then

$$\sqrt{n} \sum_{j=1}^{n-1} (\pi_{j+1} - \pi_j) B_j \leq \frac{\beta}{nv^{3/2}} \left\{ \frac{T}{4} + c\sqrt{T} + \frac{27M}{256}\left(\frac{5}{3} + \frac{5c}{4}\right) \right\}. \tag{13}$$

Here is the derivation, including the infinite tail. For $a = t - j$, centering gives $B_j = (\mathbb{E}|a - W_j| - |a|)/2 \geq 0$, $B_j \leq \sqrt{T}/2$ and $a_+ B_j \leq T/4$. The factorization

$$27x^4 - 256x + 256 = (3x - 4)^2(3x^2 + 8x + 16)$$

gives $B_j \le 27M/(256|a|^3)$ when $a \ne 0$. The exact adjacent-mass identity is

$$\pi_{j+1} - \pi_j = \frac{np - q - j}{(n+1)v}\,\mathbb{P}\{\mathrm{Bin}(n+1,p) = j+1\}.$$

Discard only its nonpositive terms. The positive numerator is at most $(a+c)_+$. Since $c \le \bar{c} < 1$, at most one mesh point lies in each of $(-c, 0)$ and $[0, 1)$; together they cost $c\sqrt{T} + T/4$. For the rest use $\sum_{i\ge1} i^{-2} < 5/3$ and $\sum_{i\ge1} i^{-3} < 5/4$. The first follows from the first five terms and an integral from 5; the second from the first two terms and an integral from 2.

To finish (13), the required mode bound holds uniformly on the entire interval $p \in [1/3, 2/3]$:

$$\max_j \mathbb{P}\{\mathrm{Bin}(M,p) = j\} < \frac{\beta}{\sqrt{Mv}} \quad (M \ge 43).$$

Indeed a mode has $z = j/M = p + \delta/M$, $|\delta| \le 2/3$ and $|1 - 2p| \le 1/3$, while $v \ge 2/9$. Thus $z(1-z) \ge v(1 - 1/M - 2/M^2) > 39v/40$. Finite Stirling and the nonpositive entropy exponent bound the mass by $(2/5)\sqrt{40/39}(516/515)/\sqrt{Mv} < \beta/\sqrt{Mv}$. Here $e^{1/516} < 516/515$, and the last comparison is a rational square. Finally $\sqrt{n}/(n+1)^{3/2} \le 1/n$ proves (13).

A positive adjacent mass requires $j < np$. Its actual variance $(n-j)s_0 + js_1$ is at most $\lambda/q \le T = \lambda/q_*$. Independence supplies $M = A^2T + 3T^2$. The affine guard gives $J(\lambda) \ge 2\sqrt{j_0 j_1 \lambda} \ge \sqrt{T}$, so all these quantities match (8).

5 Both normal signs and the cubic correction

Schulz's strict finite-binomial theorem [1, Theorem 1] states, for every $0 < p < 1$, finite n and real argument, that

$$\sqrt{n}\{F_B(j) - \Phi((j - np)/\sqrt{nv})\} < C_{\mathrm{E}}\rho_B, \qquad \rho_B = (1-2v)/\sqrt{v}.$$

Apply it at $k, k+1$ and interpolate. No sharper probability-restricted Bernoulli reserve is used. For $G(x) = \Phi((x - np)/\sqrt{nv})$,

$$G''(x) \le \frac{\phi_0(np - x)_+}{(nv)^{3/2}}.$$

The linear interpolation remainder on the whole unit interval is therefore

$$\sqrt{n}\{(1-u)G(k) + uG(k+1) - G(t)\} \le \frac{\phi_0}{2nv^{3/2}}u(1-u)(u-w)_+.$$

For $H \le 0$, changing the normal variance from v to $v + s$ is favorable. Using $u(1-u) \le 1/4$ and $u^2(1-u) \le 4/27$ gives a total normal cost at most $\phi_+(\xi/8 + 2/27)/(nv^{3/2})$. Here $c = (-q-w)_+ \le \bar{c}$. For $H \ge 0$, $c = 0$, interpolation costs at most $\phi_+(2/27)/(nv^{3/2})$, and the variance correction is at most $\phi_0 ws/(2v^{3/2})$. The inequality $1 - (1+x)^{-1/2} \le x/2$ proves the latter estimate. These are exactly the two normal terms in (8).

The conditional supports have fixed signs relative to p, since $d < r$. Expanding their cubes gives the actual raw third absolute moment

$$v(1-2v) + 3v(s_0 + s_1) - q\mathbb{E}U_0^3 + p\mathbb{E}U_1^3.$$

The inward bounds $\mathbb{E}U_0^3 \le a_0 s_0$, $\mathbb{E}U_1^3 \ge -a_1 s_1$ give an increase of at least $(3r - d)s$. Since $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$ for $x \ge 0$,

$$(v+s)^{3/2}(\rho - \rho_B) \ge s\left\{3r - d - \frac{3}{2}(1-2v) - \frac{3(1-2v)s}{8v}\right\} \ge s(K_0 - K_1 s) > 0.$$

Thus the cubic correction is favorable throughout the stated noise interval.

Combine the exact identity, (12), (13) and the two signed normal estimates. The full signed excess is strictly less than $-W(\lambda)+E_{\pm}(\lambda)/n$. Both $E_{\pm}$ are nonnegative convex quadratics. Moreover

$$W''(\lambda) = -\eta m \left\{ \frac{z^2}{4}\big[(z\lambda-1)^2+1\big] + \frac{3b^2}{8} \right\} < 0.$$

Consequently $P_{\pm}$ are strictly concave on the whole real line. Their four positive endpoint values therefore prove $P_{\pm} > 0$ throughout $[L, U]$. Since $n \geq N$ and $E_{\pm} \geq 0$, this proves Lemma 2, including both signs of H, every phase and all threshold atoms.

6 Checking the rows

The remaining parameters are in Table 2; all omitted quantities are defined by the domain row, not chosen independently.

Row	κ	ξ	c_-	ζ	m	V	$\bar{c}$	j_0	j_1
Compact	3	93/100	6	23/100	3/4	113/1000	79/200	1/5	5/2
Right	3	231/250	5	23/100	13/20	117/1000	23/50	9/40	5/2
Left	7/5	23/25	6	11/50	3/4	13/125	8/25	13/200	250/39

Table 2: Constants for the common analytic criterion.

Here is an elementary check of the only trigonometric guard. For Compact and Right, use $\theta = 2/3$ and $\sin(\pi\theta) = \sqrt{3}/2 > 17/20$. On $[0, \theta]$, concavity gives $\sin(\pi x) \geq (51/40)x$, whose squared coefficient proves $\kappa = 3$. The Compact diameter $A + d$ is below θ. For Right, on $[\theta, 84/125]$ the sine is at least $17/20 - (22/7)(2/375) > 0$; twice its square exceeds $3(84/125)^2$. For Left take $\theta = 3/4$ and $\sin(\pi\theta) = \sqrt{2}/2 > 7/10$. The chord slope is at least $14/15$. On $[\theta, 96/125]$ the lower sine bound is $563/875$. Twice these squared bounds prove $\kappa = 7/5$. At zero the comparisons are weak; evenness handles negative arguments.

Every other guard and endpoint is rational, apart from the fixed normal constants enclosed by the elementary bounds $25/8 < \pi < 22/7$. The checker reconstructs the polynomials directly from (7)–(8), verifies the completed-square identity for W'', and checks all guards before the endpoints. Conservative lower bounds for the endpoint margins are:

	$P_-(L)$	$P_-(U)$	$P_+(L)$	$P_+(U)$
Compact	.002314	.061304	.003341	.063680
Right	.000461	.000885	.004654	.011247
Left	.000055	.018585	.000897	.022487

The displayed decimals are exact rational lower targets, not floating-point proofs. No grid, fitted coefficient array or truncation of an incoming-count series occurs. This completes the proof of Theorem 1.

Scope

This note replaces the repeated crossing-loss, mass, curvature, normal and cubic arguments in the three v15 comparison notes with one criterion. It proves its local statements for every actual marginal satisfying a row. It does not acquire that row for arbitrary laws, turn a signed conclusion into an absolute one without reflected-law hypotheses, or establish a finite base. The smaller-noise comparisons and the geometric acquisition are separate joins. All previous sources and certificates are preserved.

References

[1] J. Schulz, *The optimal Berry–Esseen constant in the binomial case*, doctoral dissertation, Universität Trier, 2016, Theorem 1, printed page 1. https://d-nb.info/1197702695/34.